

**LAPORAN PROYEK AKHIR PRAKTIKUM
ALGORITMA PEMROGRAMAN LANJUT**



Disusun oleh:

Akbar Rachim (Ketua) (2509106028)

M. Riyadh Zuhair (2509106029)

Ammara Fadia Fanysarahman (2509106035)

Kelas (A2 '25)

PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

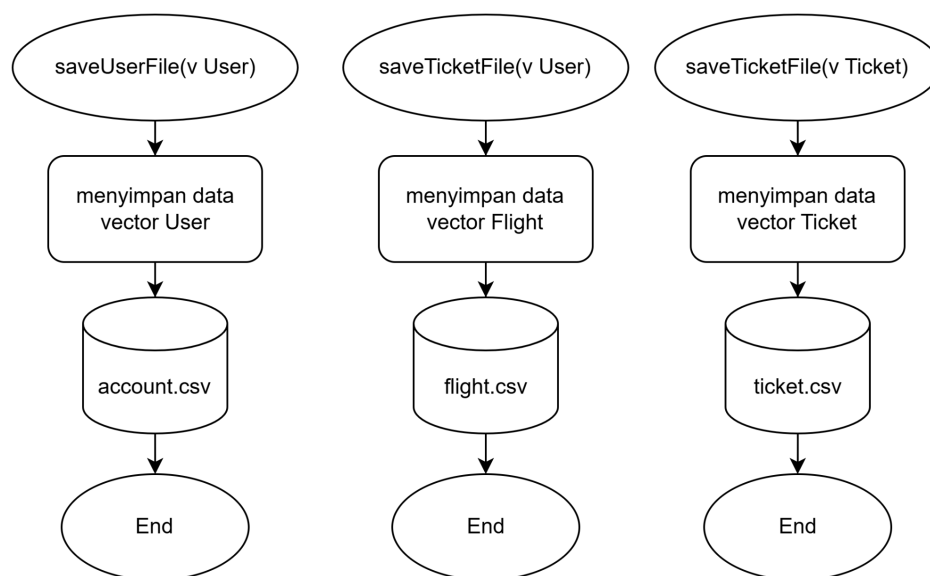
2026

1. Latar Belakang

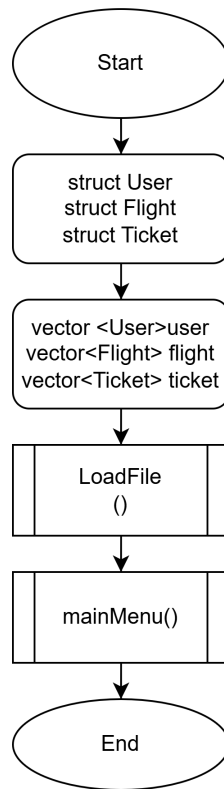
Program ini merupakan sistem manajemen tiket pesawat berbasis terminal yang dirancang untuk mengelola data pengguna, jadwal penerbangan, dan transaksi tiket secara terorganisir menggunakan struktur data *vector* di C++. Skenario penggunaan dimulai dengan pemuatan data dari file eksternal, diikuti dengan sistem autentikasi multi-user yang mencakup fitur registrasi dengan enkripsi kata sandi serta login untuk memverifikasi hak akses. Berdasarkan hasil login, akses menu akan disesuaikan secara otomatis menjadi tiga peran utama: Admin, Customer, dan Airline, di mana masing-masing peran memiliki menu khusus untuk menjalankan fungsi spesifiknya.

Program ini dikembangkan dengan pendekatan modular programming, memisahkan logika utama ke dalam berbagai fungsi khusus seperti `loginAccount()`, `registerAccount()`, dan fungsi manajemen menu untuk setiap peran guna memastikan kode yang bersih dan mudah dipelihara. Fitur-fitur utama yang tersedia meliputi manajemen akun (registrasi dan login), pemeriksaan basis data akun yang ada, enkripsi keamanan, serta sistem penyimpanan data permanen melalui fungsi `saveUserFile()`. Dengan struktur ini, program mampu menangani proses bisnis penerbangan mulai dari pengelolaan data oleh maskapai dan admin hingga pemesanan tiket oleh pelanggan secara efisien.

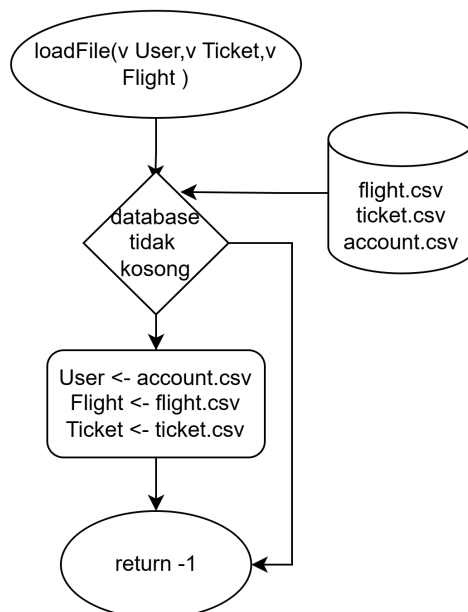
2. Flowchart



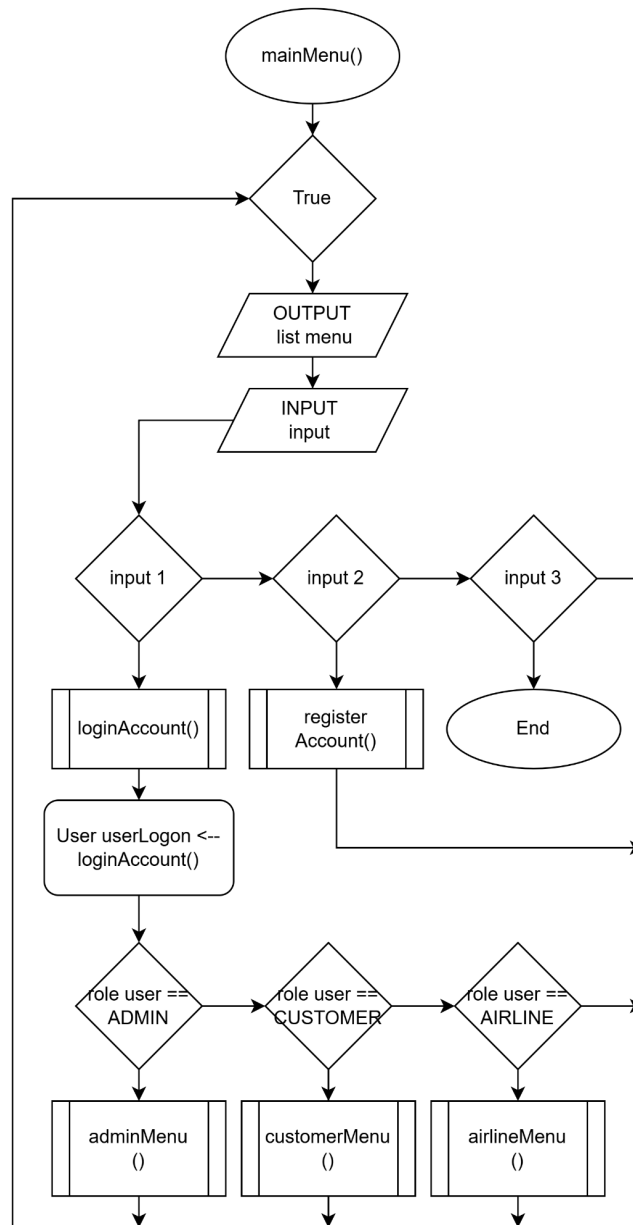
2.1 Save Data & CSV



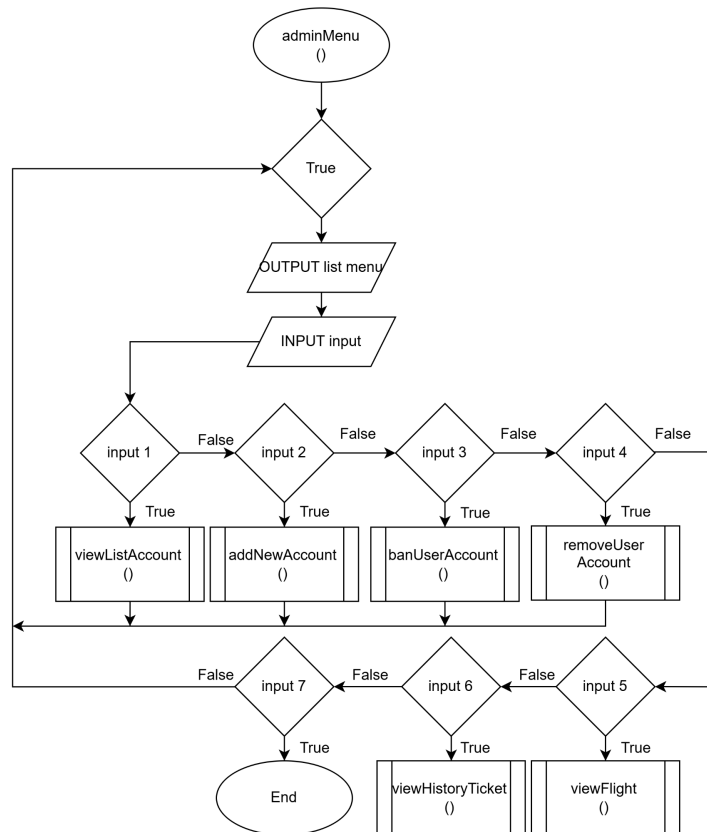
2.2 Inisialisasi



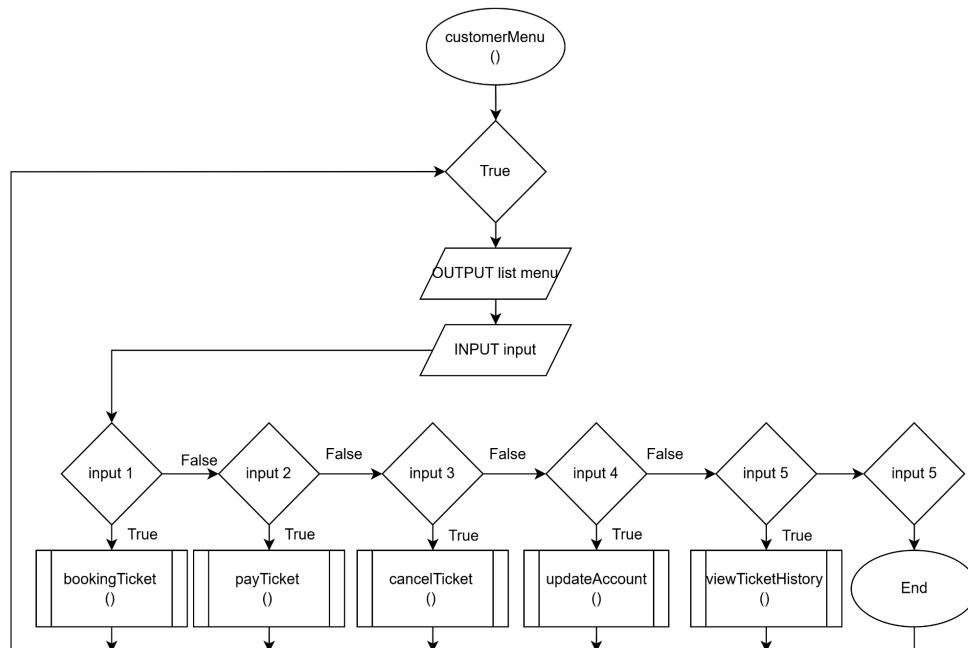
2.3 Load Data



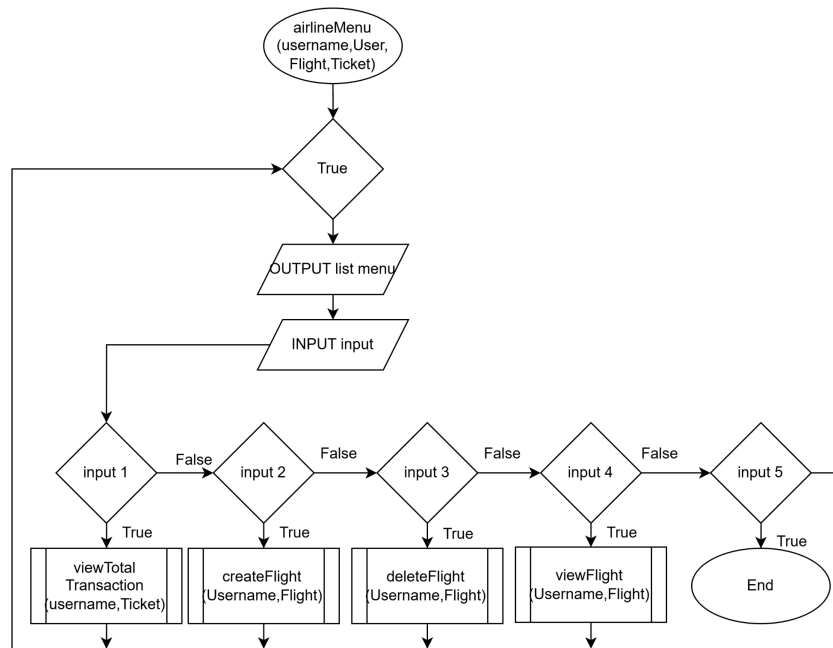
2.4 Main Menu



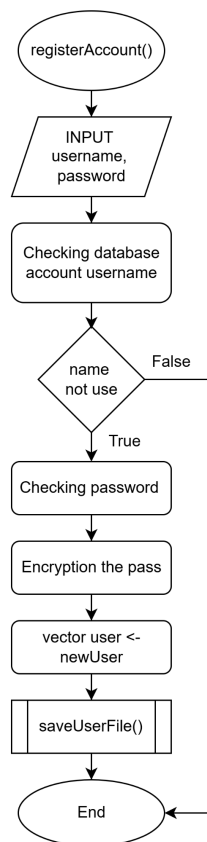
2.5 Menu Admin



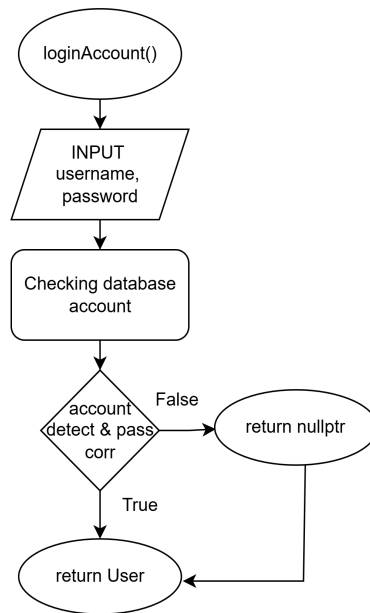
2.6 Menu User/Customer



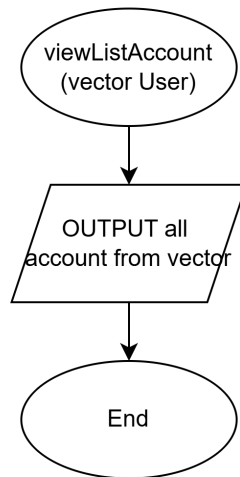
2.7 Menu Airline



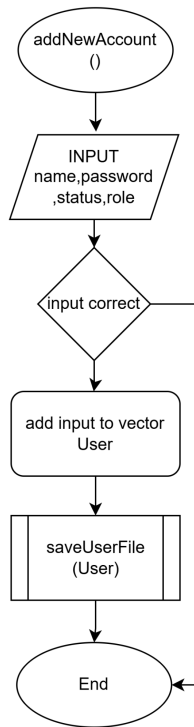
2.8 Register



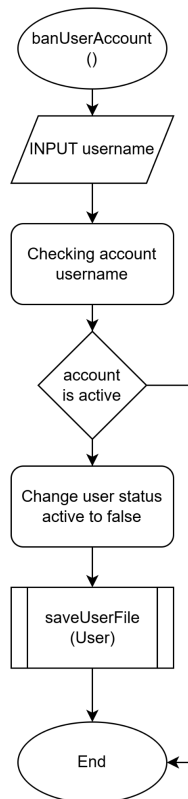
2.9 Login



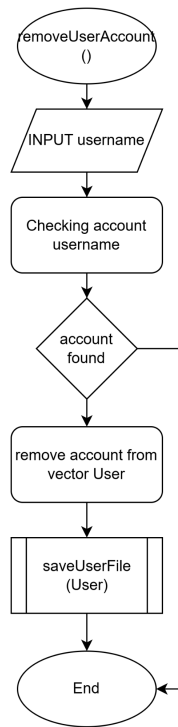
2.10 View List Account (Admin)



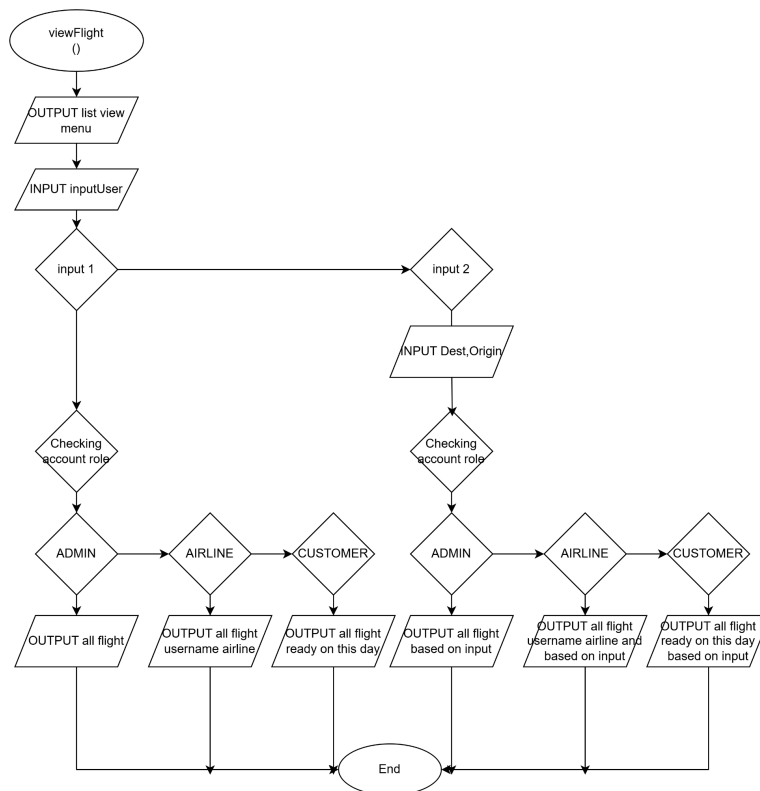
2.11 Add New Account (Admin)



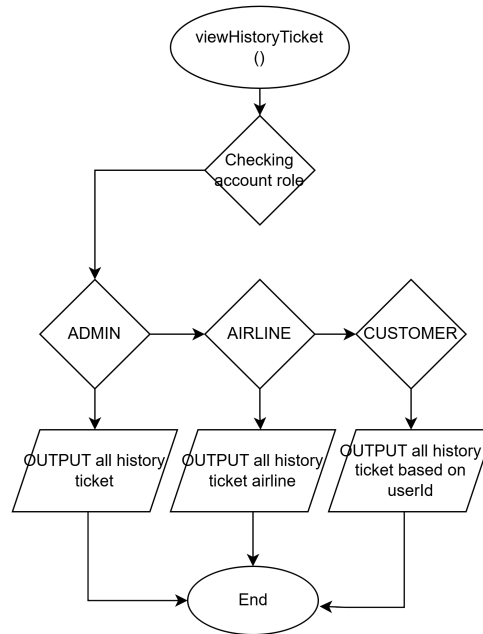
2.12 Ban User Account (Admin)



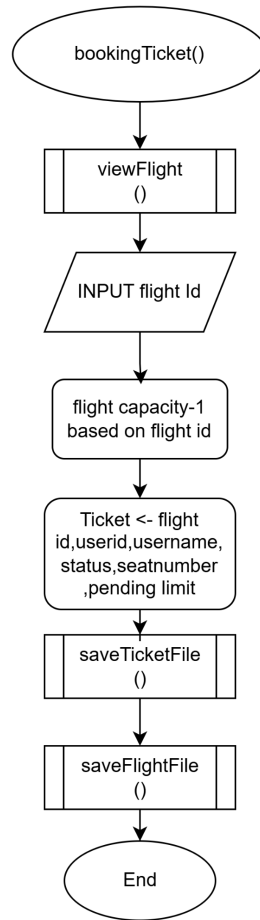
2.13 Remove User Account (Admin)



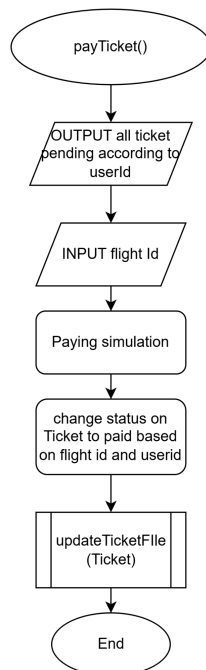
2.14 View (Flight)



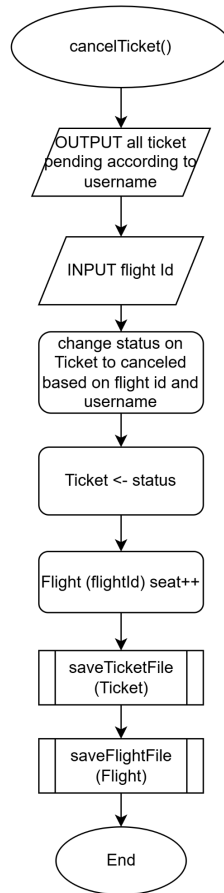
2.15 View History (Ticket)



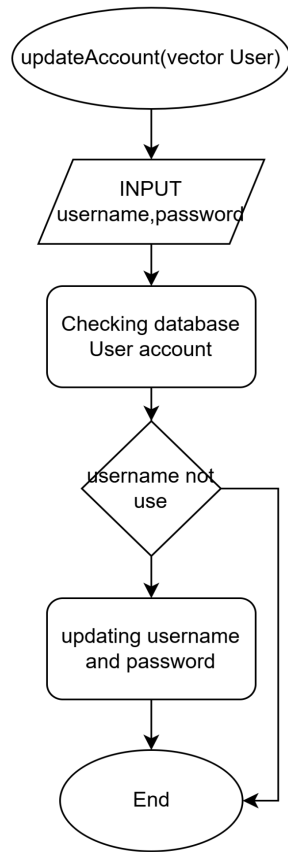
2.16 Booking (Ticket)



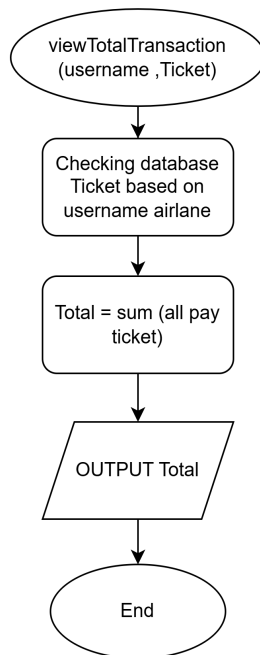
2.17 Pay (Ticket)



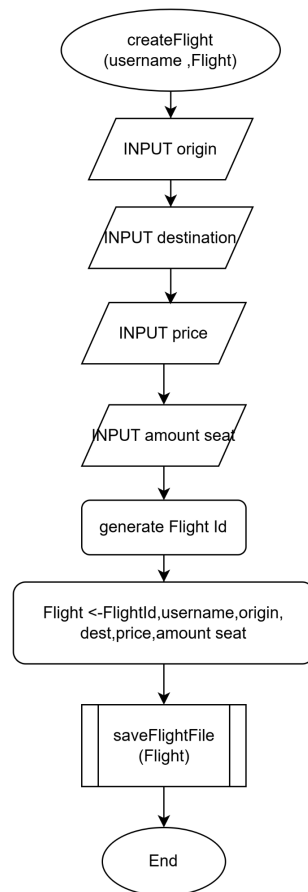
2.18 Cancel (Ticket)



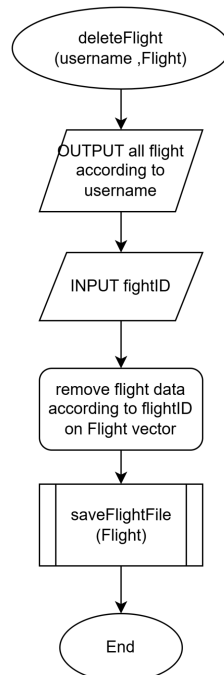
2.19 Update Account



2.20 View Total Transaction (Airline)



2.21 Create Flight (Airline)



2.22 Delete Flight (Airline)

3. Output Program

```
Greeting Admin
1. View List Account
2. Add New Account
3. Ban User Account
4. Remove User Account
5. View List Flight
6. View List Ticket Order
7. Exit
```

3.1 Login (Admin)

```
1. View Transaction
2. Add New Flight Schedule
3. Remove Flight
4. Edit Flight
5. View Flight
6. Exit
```

3.2 Login (Airline)

```
Selamat Datang Rachim
1. Booking Ticket
2. Paying Ticket
3. Cancel Ticket
4. View Ticket History
5. Exit
```

3.3 Login (User)

No	User Id	Username	Password	Account Status	Role Account
1	US1	Admin	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	ADMIN
2	US2	Rachim	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	CUSTOMER
3	US3	Batik Airline	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
4	US4	Garuda Indonesia	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
5	US5	Lion Air	ec74566fb729e8a85edafdc957a9f9bb767ca2b2074d0d661b1be051dc7678	Active	AIRLINE
6	US6	Allstar	75338b092237f438f4588e7356120be8617dd8e838666f0d9d5937c85cd9fccc	Active	AIRLINE
7	US7	Dure Airline	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
8	US8	Dubai Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE
9	US10	Ant Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE
10	US11	Code Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE

Press Enter To Continue...|

3.4 View List Account (Admin)

```

Input Name : Amengg
Input Pass : code123#
Registration Succes

```

3.6 Add New Account (Admin)

No	User Id	Username	Password	Account Status	Role Account
1	US1	Admin	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	ADMIN
2	US2	Rachim	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	CUSTOMER
3	US3	Batik Airline	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
4	US4	Garuda Indonesia	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
5	US5	Lion Air	ec74566fb729e8a85edafdc957a9f9bb767ca2b2074d0d661b1be051dc7678	Active	AIRLINE
6	US6	Allstar	75338b092237f438f4588e7356120be8617dd8e838666f0d9d5937c85cd9fccc	Active	AIRLINE
7	US7	Dure Airline	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
8	US8	Dubai Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE
9	US10	Ant Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE
10	US11	Code Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Inactive	AIRLINE
11	US12	Amengg	731bdd736ab033b2362ea223527f2c9710151398ffe11c19498ee7a753e0bf6e	Active	AIRLINE

Input User Id : US12
 Account Found : US12 | Amengg
 1. Ban Account
 2. Unban Account
 3. Exit
 Input Menu : 1
 Ban Account (y/n) : y
 Account Banned

3.7 Ban User Account (Admin)

No	User Id	Username	Password	Account Status	Role Account
1	US1	Admin	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	ADMIN
2	US2	Rachim	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	CUSTOMER
3	US3	Batik Airline	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
4	US4	Garuda Indonesia	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
5	US5	Lion Air	ec74566fb729e8a85edafdc957a9f9bb767ca2b2074d0d661b1be051dc7678	Active	AIRLINE
6	US6	Allstar	75338b092237f438f4588e7356120be8617dd8e838666f0d9d5937c85cd9fccc	Active	AIRLINE
7	US7	Dure Airline	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
8	US8	Dubai Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE
9	US10	Ant Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE
10	US11	Code Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Inactive	AIRLINE
11	US12	Amengg	731bdd736ab033b2362ea223527f2c9710151398ffe11c19498ee7a753e0bf6e	Inactive	AIRLINE

Input User Id : US12
 Account Found : US12 | Amengg
 1. Ban Account
 2. Unban Account
 3. Exit
 Input Menu : 2
 Unban Account (y/n) : y
 Account Unbanned

3.8 Unban User Account (Admin)

No	User Id	Username	Password	Account Status	Role Account
1	US1	Admin	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	ADMIN
2	US2	Rachim	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	CUSTOMER
3	US3	Batik Airline	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
4	US4	Garuda Indonesia	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
5	US5	Lion Air	ec74566fb729e8a85edafdc957a9f9bb767ca2b2074d0d661b1be051dc7678	Active	AIRLINE
6	US6	Allstar	75338b092237f438f4588e7356120be8617dd8e838666f0d9d5937c85cd9fccc	Active	AIRLINE
7	US7	Dure Airline	8d95e45ce01a698a09dea57a1250e810b7fb83f80c2b0f9eea808e80971a66c6	Active	AIRLINE
8	US8	Dubai Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE
9	US10	Ant Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Active	AIRLINE
10	US11	Code Airline	ac0e3cf6cd3517f1e32bd3909a035ccd14efab157043fe993e7663787be27c57	Inactive	AIRLINE
11	US12	Amengg	731bdd736ab033b2362ea223527f2c9710151398ffe11c19498ee7a753e0bf6e	Active	AIRLINE

Input User Id : US12
 Account Found : US12 | Amengg
 Remove Account (y/n) ?
 y
 Account Removed

3.9 Remove User Account (Admin)

No	Flight Id	Airline Name	Origin	Destination	Date Time	Price	Capacity
1	FL1	US3	Samarinda	Jakarta	2026-10-20 10:20	10000000	190
2	FL2	US3	Aceh	Jakarta	2026-10-20 10:20	100000	150
3	FL3	US3	Tarakan	Jakarta	2026-10-20 10:20	100000	120
4	FL4	US4	Maluku	Jakarta	2026-10-20 10:20	100000	129
5	FL5	US4	Samarinda	Lombok	2027-01-20 10:29	200000	50

Press Enter To Continue...

3.10 View List Flight (Admin)

```

Input Origin : Samarinda
Input Destination : Jakarta
+-----+-----+-----+-----+-----+-----+-----+-----+
| No | Flight Id | Airline Name | Origin | Destination | Date Time | Price | Capacity |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | FL1 | US3 | Samarinda | Jakarta | 2026-10-20 10:20 | 10000000 | 190 |
+-----+-----+-----+-----+-----+-----+-----+-----+
Press Enter To Continue...

```

3.11 Searching View List Flight (Admin)

```

+-----+-----+-----+-----+-----+-----+-----+-----+
| No | Ticket Id | Flight Id | User Id | Username | Ticket Status | Seat Number | Pending Limit |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | TIX1 | FL1 | US2 | Rachim | Paid | A100 | 2026-05-07 10:15 |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 2 | TIX2 | FL1 | US2 | Akbar | Paid | A99 | 2026-05-07 10:20 |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 3 | TIX3 | FL2 | US2 | Rachimm | Canceled | A150 | 2026-05-07 21:58 |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 4 | TIX7 | FL1 | US2 | Udin | Paid | A98 | 2026-05-14 17:39 |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 5 | TIX8 | FL1 | US2 | Rudi | Canceled | A97 | 2026-05-14 19:44 |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 6 | TIX9 | FL1 | US2 | Y | Canceled | A100 | 2026-05-14 20:00 |
+-----+-----+-----+-----+-----+-----+-----+-----+
Press Enter To Continue...

```

3.12 View List Ticket Order

```

Greeting Admin
1. View List Account
2. Add New Account
3. Ban User Account
4. Remove User Account
5. View List Flight
6. View List Ticket Order
7. Exit
Input : 7
1. Login
2. Register
3. Exit
Input :

```

3.13 Exit (Admin)

```

Flight FL1 | Samarinda -> Jakarta | Tickets Sold: 3 | Revenue: 30000000
Flight FL2 | Aceh -> Jakarta | Tickets Sold: 0 | Revenue: 0
Flight FL3 | Tarakan -> Jakarta | Tickets Sold: 0 | Revenue: 0
Total Revenue : 30000000
Press Enter To Continue...

```

3.14 View Transaction (Airline)

```

Input Origin : Malang
Input Destination : Balikpapan
Input Date Time (YYYY-MM-DD H:M): 2026-05-20 07:00
Input Price : 1500000
Input Capacity : 120
Added New Flight(y/n)? : y
Added Flight Succesfull

```

3.15 Add New Flight Schedule (Airline)

No	Flight Id	Airline Name	Origin	Destination	Date Time	Price	Capacity
1	FL1	US3	Samarinda	Jakarta	2026-10-20 10:20	10000000	190
2	FL2	US3	Aceh	Jakarta	2026-10-20 10:20	100000	150
3	FL3	US3	Tarakan	Jakarta	2026-10-20 10:20	100000	120
4	FL6	US3	Malang	Balikpapan	2026-05-20 07:00	1500000	120

```

Input Flight Id : FL6
Flight Found : FL6 | US3
Delete (y/n): y
Delete Flight Succesfull

```

3.16 Remove Flight (Airline)

```

Input Origin: Jakarta
Input Destination: Malang
Input Price: 290000
Input Capacity: 190

```

No	Flight Id	Airline Name	Origin	Destination	Date Time	Price	Capacity
1	FL6	US3	Jakarta	Malang	2026-10-12 10:00	290000	190

```

Input Flight Id : FL6
Flight Found : FL6 | US3
Delete (y/n): Y
Delete Flight Succesfull

```

3.17 Remove Flight Using Search (Airline)

No	Flight Id	Airline Name	Origin	Destination	Date Time	Price	Capacity
1	FL1	US3	Samarinda	Jakarta	2026-10-20 10:20	10000000	190
2	FL2	US3	Aceh	Jakarta	2026-10-20 10:20	100000	150
3	FL3	US3	Tarakan	Jakarta	2026-10-20 10:20	100000	120
4	FL6	US3	Samarinda	Malang	2026-05-19 07:00	1400000	130

```

Input Flight Id : FL6
Flight Found : FL6 | US3
Edit? (y/n): y
Input Capacity : 135
Edit Flight Succesfull

```

3.18 Edit Flight (Airline)

```

Input Origin : Samarinda
Input Destination : Jakarta
+-----+-----+-----+-----+-----+-----+-----+
| No | Flight Id | Airline Name | Origin | Destination | Date Time | Price | Capacity |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | FL1 | US3 | Samarinda | Jakarta | 2026-10-20 10:20 | 10000000 | 189 |
+-----+-----+-----+-----+-----+-----+-----+
| 2 | FL6 | US3 | Samarinda | Jakarta | 2026-05-20 07:00 | 1400000 | 120 |
+-----+-----+-----+-----+-----+-----+-----+
Input Flight Id : FL6
Flight Found : FL6 | US3
Edit? (y/n): y
Input Capacity : 130
Edit Flight Succesfull

```

3.19 Edit Flight Using Search (Airline)

```

+-----+-----+-----+-----+-----+-----+-----+
| No | Flight Id | Airline Name | Origin | Destination | Date Time | Price | Capacity |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | FL1 | US3 | Samarinda | Jakarta | 2026-10-20 10:20 | 10000000 | 190 |
+-----+-----+-----+-----+-----+-----+-----+
| 2 | FL2 | US3 | Aceh | Jakarta | 2026-10-20 10:20 | 100000 | 150 |
+-----+-----+-----+-----+-----+-----+-----+
| 3 | FL3 | US3 | Tarakan | Jakarta | 2026-10-20 10:20 | 100000 | 120 |
+-----+-----+-----+-----+-----+-----+-----+
| 4 | FL6 | US3 | Samarinda | Malang | 2026-05-19 07:00 | 1400000 | 125 |
+-----+-----+-----+-----+-----+-----+-----+
Press Enter To Continue...|

```

Gambar 3.20 View Flight (Airline)

```

1. View Transaction
2. Add New Flight Schedule
3. Remove Flight
4. Edit Flight
5. View Flight
6. Exit
Input User : 6
1. Login
2. Register
3. Exit
Input :

```

Gambar 3.21 Exit (Airline)

No	Flight Id	Airline Name	Origin	Destination	Date Time	Price	Capacity
1	FL1	Batik Airline	Samarinda	Jakarta	2026-10-20 10:20	10000000	189
2	FL2	Batik Airline	Aceh	Jakarta	2026-10-20 10:20	100000	149
3	FL3	Batik Airline	Tarakan	Jakarta	2026-10-20 10:20	100000	120
4	FL4	Garuda Indonesia	Maluku	Jakarta	2026-10-20 10:20	100000	128
5	FL5	Garuda Indonesia	Samarinda	Lombok	2027-01-20 10:29	200000	50
6	FL6	Batik Airline	Samarinda	Jakarta	2026-05-20 07:00	1400000	130

Input Flight Id
 FL6
 Flight ID : FL6
 Airline ID : US3
 Origin : Samarinda
 Destination : Jakarta
 Departure : 2026-05-20 07:00
 Price : 1400000
 Capacity : 130
 Input Passenger Name : Akbar
 Booking Ticket (y/n) : y

Gambar 3.22 Booking Ticket (User)

Input Origin : Samarinda
 Input Destination : Jakarta

No	Flight Id	Airline Name	Origin	Destination	Date Time	Price	Capacity
1	FL1	Batik Airline	Samarinda	Jakarta	2026-10-20 10:20	10000000	189
2	FL6	Batik Airline	Samarinda	Jakarta	2026-05-20 07:00	1400000	129

Input Flight Id
 FL6
 Flight ID : FL6
 Airline ID : US3
 Origin : Samarinda
 Destination : Jakarta
 Departure : 2026-05-20 07:00
 Price : 1400000
 Capacity : 129
 Input Passenger Name : Cek
 Booking Ticket (y/n) : y

Gambar 3.23 Booking Ticket Using Search (User)

No	Ticket Id	Flight Id	User Id	Username	Ticket Status	Seat Number	Pending Limit
1	TIX11	FL2	US2	Y	Pending	A150	2026-05-19 01:32
2	TIX12	FL4	US2	y	Pending	A129	2026-05-19 01:48
3	TIX13	FL6	US2	Akbar	Pending	A130	2026-05-19 02:29
4	TIX14	FL6	US2	Cek	Pending	A129	2026-05-19 02:30

Input Ticket Id : TIX13
 Pay Ticket ? (y/n) : y
 Process....

Gambar 3.24 Paying Ticket (User)

No	Ticket Id	Flight Id	Username	Ticket Status	Seat Number	Pending Limit
1	TIX11	FL2	Y	Pending	A150	2026-05-19 01:32
2	TIX12	FL4	y	Pending	A129	2026-05-19 01:48
3	TIX14	FL6	Cek	Pending	A129	2026-05-19 02:30

Input Ticket Id : TIX14
Cancel Ticket ? (y/n) : y
Process....

Gambar 3.25 Cancel Ticket (User)

No	Ticket Id	Flight Id	User Id	Username	Ticket Status	Seat Number	Pending Limit
1	TIX1	FL1	US2	Rachim	Paid	A100	2026-05-07 10:15
2	TIX2	FL1	US2	Akbar	Paid	A99	2026-05-07 10:20
3	TIX3	FL2	US2	Rachimm	Canceled	A150	2026-05-07 21:58
4	TIX7	FL1	US2	Udin	Paid	A98	2026-05-14 17:39
5	TIX8	FL1	US2	Rudi	Canceled	A97	2026-05-14 19:44
6	TIX9	FL1	US2	Y	Canceled	A100	2026-05-14 20:00
7	TIX10	FL1	US2	Y	Paid	A190	2026-05-19 01:27
8	TIX11	FL2	US2	Y	Canceled	A150	2026-05-19 01:32
9	TIX12	FL4	US2	y	Pending	A129	2026-05-19 01:48
10	TIX13	FL6	US2	Akbar	Paid	A130	2026-05-19 02:29
11	TIX14	FL6	US2	Cek	Canceled	A129	2026-05-19 02:30

Press Enter To Continue...

Gambar 3.26 View Ticket History (User)

4. Langkah-langkah Git

```
> ~/Documents/Tugas
bero@bero-Aspire-AL14-71P:~/Documents/Tugas$ git clone https://github.com/cybercrym-crying/Manajemen-Tiket-Pesawat.git
Cloning into 'Manajemen-Tiket-Pesawat'...
remote: Enumerating objects: 437, done.
remote: Counting objects: 100% (190/190), done.
remote: Compressing objects: 100% (126/126), done.
remote: Total 437 (delta 96), reused 126 (delta 64), pack-reused 247 (from 1)
Receiving objects: 100% (437/437), 6.79 MiB | 3.34 MiB/s, done.
Resolving deltas: 100% (239/239), done.
```

Gambar 4.1 Git Clone

```
D:\pa-apl\Manajemen-Tiket-Pesawat>git pull origin main
From https://github.com/ammaraFadia/Manajemen-Tiket-Pesawat
 * branch                main          -> FETCH_HEAD
Updating b995827..1f3c86d
Fast-forward
 data/flight.csv |      3 ++-
 data/ticket.csv |      2 +-
 program         | Bin 737584 -> 737584 bytes
 program.exe     | Bin 4102615 -> 4067943 bytes
 src/flight.cpp  |      1 +
 src/utils.cpp   |      3 +++
 6 files changed, 7 insertions(+), 2 deletions(-)
```

Gambar 4.2 Git Pull

```
bero@bero-Aspire-AL14-71P:~/Documents/Tugas/Manajemen-Tiket-Pesawat$ git add .
```

Gambar 4.3 Git Add

```
bero@bero-Aspire-AL14-71P:~/Documents/Tugas/Manajemen-Tiket-Pesawat$ git commit -m "fix output"
[main 428d395] fix output
 4 files changed, 4 insertions(+), 4 deletions(-)
```

Gambar 4.4 Git Commit

```
bero@bero-Aspire-AL14-71P:~/Documents/Tugas/Manajemen-Tiket-Pesawat$ git push -u origin main
Username for 'https://github.com': cybercrym-crying
Password for 'https://cybercrym-crying@github.com':
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 729 bytes | 729.00 KiB/s, done.
Total 9 (delta 7), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (7/7), completed with 7 local objects.
To https://github.com/cybercrym-crying/Manajemen-Tiket-Pesawat.git
 1f3c86d..428d395  main -> main
branch 'main' set up to track 'origin/main'.
```

Gambar 4.5 Git Push

5. Deskripsi Pembagian Tugas

- Akbar Rachim : Program
- M. Riyadh Zuhair : Error Handling
- Ammara Fadia F. : Menemukan dan memperbaiki bug